

12. Di penny tambahkan

Nomor 12: basic auth untuk /admin

PASS="pakar_pinter_jadi_gob***"

htpasswd -bc /etc/apache2/.htpasswd prabs "\$PASS"

<Location /admin>

AuthType Basic

AuthName "Area Terbatas"

AuthUserFile /etc/apache2/.htpasswd

Require valid-user

</Location>

Test

curl -i http://www.k36.com/admin/

curl -i -u 'prabs:pakar_pinter_jadi_gob***' http://www.k36.com/admin/

```
root@alpha:~# curl -i http://www.k36.com/admin/
HTTP/1.1 401 Unauthorized
Date: Thu, 01 Oct 2026 11:03:57 GMT
Server: Apache/2.4.68 (Debian)
WWW-Authenticate: Basic realm="Area Terbatas"
Content-Length: 498
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at www.k36.com Port 80</address>
</body></html>
root@alpha:~# curl -i -u 'prabs:pakar_pinter_jadi_gob***' http://www.k36.com/admin/
HTTP/1.1 404 Not Found
Date: Thu, 01 Oct 2026 11:04:05 GMT
Server: Apache/2.4.68 (Debian)
Content-Length: 313
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL was not found on this server.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at www.k36.com Port 80</address>
</body></html>
```

```

root@alpha:~# curl -i -u 'prabs:salah' http://www.k36.com/admin/
HTTP/1.1 401 Unauthorized
Date: Thu, 01 Oct 2026 11:04:13 GMT
Server: Apache/2.4.68 (Debian)
WWW-Authenticate: Basic realm="Area Terbatas"
Content-Length: 498
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.68 (Debian) Server at www.k36.com Port 80</address>
</body></html>

```

13. Di penny (apache)

```
cat <<'EOF' > /etc/apache2/sites-available/000-redirect.conf
```

```
<VirtualHost *:80>
```

```
    ServerName penny.k36.com
```

```
    ServerAlias 192.229.4.2
```

```
    Redirect permanent / http://www.k36.com/
```

```
</VirtualHost>
```

```
EOF
```

```
a2dissite 000-default.conf
```

Di abbey (nginx)

```
server {
```

```
    listen 80;
```

```
    server_name abbey.k36.com 192.229.2.2;
```

```
    return 302 http://static.k36.com$request_uri;
```

```
}
```

Pengecekan dari alpha

```
curl -I http://192.229.4.2
curl -I http://penny.k36.com
curl -IL http://penny.k36.com
curl -I http://192.229.2.2
curl -I http://abbey.k36.com
curl -IL http://abbey.k36.com
curl -I http://www.k36.com
curl -I http://static.k36.com
```

Output

```
root@alpha:~# curl -I http://192.229.4.2
HTTP/1.1 301 Moved Permanently
Date: Wed, 30 Sep 2026 14:50:11 GMT
Server: Apache/2.4.68 (Debian)
Location: http://www.k36.com/
Content-Type: text/html; charset=iso-8859-1
```

```
root@alpha:~# curl -I http://penny.k36.com
HTTP/1.1 301 Moved Permanently
Date: Wed, 30 Sep 2026 14:50:17 GMT
Server: Apache/2.4.68 (Debian)
Location: http://www.k36.com/
Content-Type: text/html; charset=iso-8859-1
```

```
root@alpha:~# curl -I http://www.k36.com
HTTP/1.1 200 OK
Date: Wed, 30 Sep 2026 14:50:23 GMT
Server: Apache/2.4.68 (Debian)
Last-Modified: Wed, 30 Sep 2026 14:07:34 GMT
ETag: "29cf-65cb3d2275fca"
Accept-Ranges: bytes
Content-Length: 10703
Vary: Accept-Encoding
Content-Type: text/html
```

```
root@alpha:~# curl -IL http://penny.k36.com
HTTP/1.1 301 Moved Permanently
Date: Wed, 30 Sep 2026 14:50:28 GMT
Server: Apache/2.4.68 (Debian)
Location: http://www.k36.com/
Content-Type: text/html; charset=iso-8859-1

HTTP/1.1 200 OK
Date: Wed, 30 Sep 2026 14:50:28 GMT
Server: Apache/2.4.68 (Debian)
Last-Modified: Wed, 30 Sep 2026 14:07:45 GMT
ETag: "29cf-65cb3d2c5c884"
Accept-Ranges: bytes
Content-Length: 10703
Vary: Accept-Encoding
Content-Type: text/html
```

```
root@alpha:~# curl -I http://www.k36.com
HTTP/1.1 200 OK
Date: Wed, 30 Sep 2026 15:07:28 GMT
Server: Apache/2.4.68 (Debian)
Last-Modified: Wed, 30 Sep 2026 14:07:34 GMT
ETag: "29cf-65cb3d2275fca"
Accept-Ranges: bytes
Content-Length: 10703
Vary: Accept-Encoding
Content-Type: text/html

root@alpha:~# curl -I http://static.k36.com
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 30 Sep 2026 15:07:39 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive
```

```
root@alpha:~# curl -I http://192.229.2.2
HTTP/1.1 302 Moved Temporarily
Server: nginx
Date: Wed, 30 Sep 2026 15:00:31 GMT
Content-Type: text/html
Content-Length: 138
Connection: keep-alive
Location: http://static.k36.com/

root@alpha:~# curl -I http://abbey.k36.com
HTTP/1.1 302 Moved Temporarily
Server: nginx
Date: Wed, 30 Sep 2026 15:00:47 GMT
Content-Type: text/html
Content-Length: 138
Connection: keep-alive
Location: http://static.k36.com/

root@alpha:~#
root@alpha:~# curl -IL http://abbey.k36.com
HTTP/1.1 302 Moved Temporarily
Server: nginx
Date: Wed, 30 Sep 2026 15:00:59 GMT
Content-Type: text/html
Content-Length: 138
Connection: keep-alive
Location: http://static.k36.com/

HTTP/1.1 200 OK
Server: nginx
Date: Wed, 30 Sep 2026 15:00:59 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive
```

14. Di obladi dan di desmond
a2enmod remoteip

```
cat <<'EOF' > /etc/apache2/conf-available/remoteip.conf
```

```
RemoteIPHeader X-Forwarded-For
```

```
RemoteIPInternalProxy 192.229.4.2
```

```
EOF
```

```
a2enconf remoteip
```

```
CustomLog ${APACHE_LOG_DIR}/access.log combined
```

di oblada dan molly: di server block

```
set_real_ip_from 192.229.2.2;
```

```
real_ip_header X-Real-IP;
```

Pengecekan di alpha

Apache for i in 1 2 3 4 5 6; do curl -s -o /dev/null http://www.k36.com/; done

Nginx for i in 1 2 3 4 5 6; do curl -s -o /dev/null http://static.k36.com/; done

```
192.229.3.2 - - [30/Sep/2026:19:38:24 +0000] "HEAD / HTTP/1.1" 200 311 "-" "curl/8.14.1"
root@obladi:~# tail -n 3 /var/log/apache2/access.log
192.229.3.2 - - [30/Sep/2026:19:42:11 +0000] "GET / HTTP/1.1" 200 11014 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:42:12 +0000] "GET / HTTP/1.1" 200 11013 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:42:12 +0000] "GET / HTTP/1.1" 200 11013 "-" "curl/8.14.1"
root@obladi:~#
```

```
invoke-rc.d: policy-rc.d denied execution of restart.
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
Restarting nginx: nginx.
88888888b.      888      d8b 888b      888      888
888 "Y88b      888      Y8P 8888b      888      888
888 888      888      88888b 888      888
888 888 .d88b. 88888b. 888 888Y88b 888 .d88b. 888888
888 888 d8P Y8b 888 "88b 888 888 Y88b888 d8P Y8b 888
888 888 88888888 888 888 888 888 Y88888 888888888 888
888 .d88P Y8b.      888 d88P 888 888 Y8888 Y8b.      Y88b.
88888888P" "Y8888 88888P" 888 888 Y888 "Y8888 "Y888
```

```
DebiNet - Lightweight Debian-based Networking Toolbox
Type "debinet-tools" for available utilities
```

```
root@molly:~# tail -n 5 /var/log/nginx/access.log
192.229.3.2 - - [30/Sep/2026:19:38:54 +0000] "GET / HTTP/1.0" 200 30 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:38:54 +0000] "GET / HTTP/1.0" 200 30 "-" "curl/8.14.1"
root@molly:~#
```

```

nginx: configuration file /etc/nginx/nginx.conf test is successful
Restarting nginx: nginx.
88888888b.      888      d8b 888b      888      888
888  "Y88b      888      Y8P 8888b      888      888
888  888      888      88888b 888      888
888  888 .d88b. 88888b. 888 888Y88b 888 .d88b. 888888
888  888 d8P  Y8b 888 "88b 888 888 Y88b888 d8P  Y8b 888
888  888 88888888 888 888 888 888 Y88888 888888888 888
888 .d88P Y8b.   888 d88P 888 888 Y8888 Y8b.   Y88b.
88888888P"  "Y8888 88888P" 888 888 Y888 "Y8888 "Y888

DebiNet - Lightweight Debian-based Networking Toolbox
Type "debinet-tools" for available utilities
root@oblada:~# tail -n 5 /var/log/nginx/access.log
192.229.3.2 - - [30/Sep/2026:19:38:03 +0000] "GET / HTTP/1.0" 200 31 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:38:54 +0000] "GET / HTTP/1.0" 200 31 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:38:54 +0000] "GET / HTTP/1.0" 200 31 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:38:54 +0000] "GET / HTTP/1.0" 200 31 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:38:54 +0000] "GET / HTTP/1.0" 200 31 "-" "curl/8.14.1"
root@oblada:~#

Type "debinet-tools" for available utilities
root@desmond:~# tail -n 3 /var/log/apache2/access.log
192.229.3.2 - - [30/Sep/2026:19:42:11 +0000] "GET / HTTP/1.1" 200 11014 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:42:11 +0000] "GET / HTTP/1.1" 200 11013 "-" "curl/8.14.1"
192.229.3.2 - - [30/Sep/2026:19:42:12 +0000] "GET / HTTP/1.1" 200 11013 "-" "curl/8.14.1"
root@desmond:~#

```

15.

Penny

```
mkdir -p /var/www/eternal
```

```
cat <<'EOF' > /var/www/eternal/index.php
```

```
<?php
```

```
echo "Eternal dari penny, PHP " . phpversion();
```

```
?>
```

```
EOF
```

```
grep -q "Listen 8081" /etc/apache2/ports.conf || echo "Listen 8081" >> /et>
```

```
cat <<'EOF' > /etc/apache2/sites-available/eternal.conf
```

```
<VirtualHost *:8081>
```

```
    DocumentRoot /var/www/eternal
```

```
</VirtualHost>
```

```
EOF
```

Di penny-proxy.conf tambahkan

```
#Nomor 15
```

```
ProxyPass /eternal http://127.0.0.1:8081
```

```
ProxyPassReverse /eternal http://127.0.0.1:8081
```

Abbey

```
#Nomor 15
```

```
mkdir -p /var/www/orion
```

```
echo "Orion dari abbey" > /var/www/orion/index.html
```

```
echo '<?php echo "x"; ?>' > /var/www/orion/test.php
```

```
#Nomor 15
```

```
location = /orion {
```

```
    return 301 /orion/;
```

```
}
```

```
location /orion/ {
```

```
    alias /var/www/orion/;
```

```
}
```

```
}
```

Output yang diharapkan

```
root@alpha:~# curl http://www.k36.com/eternal/  
Eternal dari penny, PHP 8.4.26root@alpha:~#  
root@alpha:~# curl http://www.k36.com/eternal/  
Eternal dari penny, PHP 8.4.26root@alpha:~#  
root@alpha:~# curl http://static.k36.com/orion/  
Orion dari abbey  
root@alpha:~# curl http://static.k36.com/orion/test.php  
<?php echo "x"; ?>  
root@alpha:~# curl -I http://www.k36.com/  
HTTP/1.1 200 OK  
Date: Wed, 30 Sep 2026 20:45:21 GMT  
Server: Apache/2.4.68 (Debian)  
Last-Modified: Wed, 30 Sep 2026 19:37:24 GMT  
ETag: "29cf-65cb86daf87a7"  
Accept-Ranges: bytes  
Content-Length: 10703  
Vary: Accept-Encoding  
Content-Type: text/html  
  
root@alpha:~# █
```

16.

apt update

apt install -y apache2-utils

ab -n 250 -c 10 http://www.k36.com/

ab -n 250 -c 10 <http://static.k36.com/>

```
root@alpha:~# ab -n 250 -c 10 http://www.k36.com/
This is ApacheBench, Version 2.3 <$Revision: 1934973 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

```
Benchmarking www.k36.com (be patient)
Completed 100 requests
Completed 200 requests
Finished 250 requests
```

```
Server Software:      Apache/2.4.68
Server Hostname:      www.k36.com
Server Port:          80
```

```
Document Path:        /
Document Length:       10703 bytes
```

```
Concurrency Level:     10
Time taken for tests:   0.142 seconds
Complete requests:     250
Failed requests:        0
Total transferred:     2744250 bytes
HTML transferred:      2675750 bytes
Requests per second:   1758.87 [#/sec] (mean)
Time per request:      5.685 [ms] (mean)
Time per request:      0.569 [ms] (mean, across all concurrent requests)
Transfer rate:         18854.57 [Kbytes/sec] received
```

Connection Times (ms)

	min	mean[+/-sd]	median	max
Connect:	0	1 0.6	1	4
Processing:	2	4 1.2	4	9
Waiting:	2	4 1.2	3	9
Total:	3	5 1.2	5	10

Percentage of the requests served within a certain time (ms)

50%	5
66%	5
75%	6
80%	6
90%	7
95%	8
98%	9
99%	9
100%	10 (longest request)

```
root@alpha:~# ab -n 250 -c 10 http://static.k36.com/
This is ApacheBench, Version 2.3 <$Revision: 1934973 $>
```

```
100%      10 (longest request)
root@alpha:~# ab -n 250 -c 10 http://static.k36.com/
This is ApacheBench, Version 2.3 <$Revision: 1934973 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

```
Benchmarking static.k36.com (be patient)
Completed 100 requests
Completed 200 requests
Finished 250 requests
```

```
Server Software:      nginx
Server Hostname:      static.k36.com
Server Port:          80
```

```
Document Path:        /
Document Length:       31 bytes
```

```
Concurrency Level:     10
Time taken for tests:   0.100 seconds
Complete requests:      250
Failed requests:        124
    (Connect: 0, Receive: 0, Length: 124, Exceptions: 0)
Total transferred:     40126 bytes
HTML transferred:       7626 bytes
Requests per second:    2497.83 [#/sec] (mean)
Time per request:       4.003 [ms] (mean)
Time per request:       0.400 [ms] (mean, across all concurrent requests)
Transfer rate:          391.51 [Kbytes/sec] received
```

Connection Times (ms)

	min	mean[+/-sd]	median	max
Connect:	0	1 0.3	1	2
Processing:	2	3 1.0	3	8
Waiting:	2	3 1.0	3	8
Total:	3	4 1.0	4	9

Percentage of the requests served within a certain time (ms)

50%	4
66%	4
75%	4
80%	4
90%	5
95%	6
98%	8
99%	9
100%	9 (longest request)

17.

Di prab

@ IN SOA prab.k36.com. root.k36.com. (2026100101 604800 86400
2419200 604800)

alpha IN TXT "alpha"

beta IN TXT "beta"

gamma IN TXT "gamma"

delta IN TXT "delta"

epsilon IN TXT "epsilon"

```

root@alpha:~# dig @192.229.1.3 beta.k36.com TXT

;<<>> DiG 9.20.26-1~deb13u1-Debian <<>> @192.229.1.3 beta.k36.com TXT
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 14090
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: c2ac674c5c0da85601000006abd8b000b5d3d1b8f87f7bb (good)
;; QUESTION SECTION:
;beta.k36.com.                IN      TXT

;; ANSWER SECTION:
beta.k36.com.                604800 IN      TXT      "beta"

;; Query time: 0 msec
;; SERVER: 192.229.1.3#53(192.229.1.3) (UDP)
;; WHEN: Wed Sep 30 22:19:44 UTC 2026
;; MSG SIZE rcvd: 86

```

```

root@alpha:~# dig @192.229.1.2 alpha.k36.com TXT

;<<>> DiG 9.20.26-1~deb13u1-Debian <<>> @192.229.1.2 alpha.k36.com TXT
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 28934
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 9f566791fd4c997e01000006abd8aedf0c3b92f561fbabb (good)
;; QUESTION SECTION:
;alpha.k36.com.                IN      TXT

;; ANSWER SECTION:
alpha.k36.com.                604800 IN      TXT      "alpha"

;; Query time: 0 msec
;; SERVER: 192.229.1.2#53(192.229.1.2) (UDP)
;; WHEN: Wed Sep 30 22:19:25 UTC 2026
;; MSG SIZE rcvd: 88

```

```

root@alpha:~# dig @192.229.1.2 beta.k36.com TXT

;<<>> DiG 9.20.26-1~deb13u1-Debian <<>> @192.229.1.2 beta.k36.com TXT
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31682
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: a9c09100d1dcef4101000006abd8af7f77c1bda9129de3 (good)
;; QUESTION SECTION:
;beta.k36.com.                IN      TXT

;; ANSWER SECTION:
beta.k36.com.                604800 IN      TXT      "beta"

;; Query time: 0 msec
;; SERVER: 192.229.1.2#53(192.229.1.2) (UDP)
;; WHEN: Wed Sep 30 22:19:35 UTC 2026
;; MSG SIZE rcvd: 86

```

```

root@alpha:~# dig +short @192.229.1.2 k36.com SOA
dig +short @192.229.1.3 k36.com SOA
prab.k36.com. root.k36.com. 2026100101 604800 86400 2419200 604800
prab.k36.com. root.k36.com. 2026100101 604800 86400 2419200 604800
root@alpha:~# for h in alpha beta gamma delta epsilon; do echo -n "$h -> "; dig +short $h.k36.com
TXT; done
alpha -> "alpha"
beta -> "beta"
gamma -> "gamma"
delta -> "delta"
epsilon -> "epsilon"
root@alpha:~#

```

18.

Di prab membuat [soal18.sh](#)

```
cat <<'EOF' > /root/soal18.sh
```

```
#!/bin/bash
```

```
ZONE=/etc/bind/k36/k36.com
```

```
serial() { grep -m1 -oE '\( *[0-9]+' $ZONE | grep -oE '[0-9]+'; }
```

```
setserial() { sed -E -i "/SOA/ s/\( *[0-9]+ /( $1 /" $ZONE; }
```

```
reload() {
```

```
    named-checkzone k36.com $ZONE > /dev/null || { echo "Zona error,  
    dihentikan"; exit 1; }
```

```
    service named restart > /dev/null  
}
```

```
NEW=$(( $(serial) + 1 ))
```

```
case "$1" in
```

```
ttl)
```

```
    sed -E -i "s/^abbey[[:space:]].*/abbey 15 IN A 192.229.2.2/" $ZONE
```

```
    setserial $NEW
```

```
    reload
```

```
    echo "TTL 15 aktif, serial $NEW"
```

```
;;
```

```
swap)
```

```
    echo "Tukar IP jam: $(date +%T)"
```

```
    sed -E -i "s/^abbey[[:space:]].*/abbey 15 IN A 10.77.77.77/" $ZONE
```

```
    setserial $NEW
```

```
    reload
```

```
    echo "IP fiktif aktif, serial $NEW"
```

```
;;
```

```
restore)
```

```

sed -E -i "s/^abbey[[:space:]].*/abbey IN A 192.229.2.2/" $ZONE
setserial $NEW
reload
sed -E -i "s/\\( *20261[0-9]+ /( $NEW /" /root/init.sh
echo "Kembali normal, serial $NEW (init.sh disamakan)"
;;
*)
echo "Pakai: bash /root/soal18.sh ttl|swap|restore"
;;
esac
EOF

```

Pada alpha

Buat resolver [18.sh](#)

```

export DEBIAN_FRONTEND=noninteractive
apt update && apt install -y bind9 bind9-utils

```

```

cat <<'EOF' > /etc/bind/named.conf.options
options {
    directory "/var/cache/bind";
    forwarders { 192.229.1.2; };
    forward only;
    dnssec-validation no;
    listen-on { 127.0.0.1; };
    allow-query { localhost; };
};
EOF

```

```

service named restart

```

```

dig +noall +answer @127.0.0.1 abbey.k36.com

```

Caranya: `./soal18.sh` ttl di prab, setelah itu pastikan resolver pada alpha sudah berjalan, kemudian jalankan

```
ndc flush; for i in $(seq 1 15); do echo -n "$(date +%T) "; dig +noall +answer @1
27.0.0.1 abbey.k36.com; sleep 3; done
```

Sambil menunggu jalankan `./soal18.sh swap`

```
root@prab:~# bash /root/soal18.sh swap
Tukar IP jam: 10:17:03
IP fiktif aktif, serial 2026100301
root@prab:~#
```

Terlihat terdapat cache yang masih muncul ketika swap sudah berhasil diganti pada menit ketiga, sedangkan menit kelima masih menggunakan IP yang lama

```
root@alpha:~# rndc flush; for i in $(seq 1 15); do echo -n "$(date +%T) "; dig +noall +answer @1
27.0.0.1 abbey.k36.com; sleep 3; done
10:16:53 abbey.k36.com. 15 IN A 192.229.2.2
10:16:56 abbey.k36.com. 12 IN A 192.229.2.2
10:16:59 abbey.k36.com. 9 IN A 192.229.2.2
10:17:02 abbey.k36.com. 6 IN A 192.229.2.2
10:17:05 abbey.k36.com. 3 IN A 192.229.2.2
10:17:08 abbey.k36.com. 15 IN A 10.77.77.77
10:17:11 abbey.k36.com. 12 IN A 10.77.77.77
10:17:14 abbey.k36.com. 9 IN A 10.77.77.77
10:17:18 abbey.k36.com. 5 IN A 10.77.77.77
10:17:21 abbey.k36.com. 2 IN A 10.77.77.77
10:17:24 abbey.k36.com. 12 IN A 10.77.77.77
10:17:27 abbey.k36.com. 9 IN A 10.77.77.77
10:17:30 abbey.k36.com. 6 IN A 10.77.77.77
10:17:33 abbey.k36.com. 3 IN A 10.77.77.77
10:17:36 abbey.k36.com. 15 IN A 10.77.77.77
root@alpha:~# dig +short @192.229.1.2 k36.com SOA
prab.k36.com. root.k36.com. 2026100301 604800 86400 2419200 604800
root@alpha:~# dig +short @192.229.1.3 k36.com SOA
prab.k36.com. root.k36.com. 2026100301 604800 86400 2419200 604800
root@alpha:~# dig +noall +answer @192.229.1.3 abbey.k36.com
abbey.k36.com. 15 IN A 10.77.77.77
root@alpha:~# dig +noall +answer @192.229.1.2 abbey.k36.com
abbey.k36.com. 15 IN A 10.77.77.77
root@alpha:~#
```

Jalankan `./soal18.sh restore`

Dan cek di **alpha** dengan

```
dig +short @192.229.1.2 k36.com SOA
```

dig +short @192.229.1.3 k36.com SOA

dig +noall +answer @192.229.1.2 abbey.k36.com

dig +noall +answer @192.229.1.3 abbey.k36.com

Pastikan mempunyai output yang sama

19.

Tambahkan di **prab dan tedd** pada **named.conf.options**

```
allow-recursion { any; };
```

```
allow-query { any; };
```

Di prab tambahkan

```
outbound IN CNAME http.badssl.com
```

Tes dari alpha

dig outbound.k36.com

dig @192.229.1.3 outbound.k36.com

dig +short @192.229.1.2 k36.com SOA

dig +short @192.229.1.3 k36.com SOA

```
curl -i -H "Host: http.badssl.com" http://outbound.k36.com
```

```
curl -i http://http.badssl.com
```

Yang diharapkan:

- dig menampilkan `outbound.k36.com. CNAME http.badssl.com.` lalu satu baris `A` untuk `http.badssl.com.` Kalau cuma `CNAME` tanpa baris `A`, recursion belum jalan.
- Dua serial sama (2026100399).
- Isi `curl` ke `outbound.k36.com` sama dengan isi `curl` ke `http.badssl.com`.

```

root@alpha:~# dig outbound.k36.com

; <<>> DiG 9.20.29-1~deb13u1-Debian <<>> outbound.k36.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 48161
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: fe1e06ecd23bc30b010000006abe379004b98de3e91102b6 (good)
;; QUESTION SECTION:
;outbound.k36.com.                IN      A

;; ANSWER SECTION:
outbound.k36.com.                604800  IN      CNAME   http.badssl.com.
http.badssl.com.                293    IN      A       104.154.89.105

;; Query time: 14 msec
;; SERVER: 192.229.1.2#53(192.229.1.2) (UDP)
;; WHEN: Thu Oct 01 10:36:00 UTC 2026
;; MSG SIZE rcvd: 115

```

```

root@alpha:~# dig @192.229.1.3 outbound.k36.com

; <<>> DiG 9.20.29-1~deb13u1-Debian <<>> @192.229.1.3 outbound.k36.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 55463
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 7524f7700fcff044010000006abe37960b9effc5e3f537ac (good)
;; QUESTION SECTION:
;outbound.k36.com.                IN      A

;; ANSWER SECTION:
outbound.k36.com.                604800  IN      CNAME   http.badssl.com.
http.badssl.com.                287    IN      A       104.154.89.105

;; Query time: 0 msec
;; SERVER: 192.229.1.3#53(192.229.1.3) (UDP)
;; WHEN: Thu Oct 01 10:36:06 UTC 2026
;; MSG SIZE rcvd: 115

```

```

root@alpha:~# dig +short @192.229.1.2 k36.com SOA
prab.k36.com. root.k36.com. 2026100310 604800 86400 2419200 604800
root@alpha:~# dig +short @192.229.1.3 k36.com SOA
prab.k36.com. root.k36.com. 2026100310 604800 86400 2419200 604800

```

```
root@alpha:~# curl -i http://http.badssl.com
HTTP/1.1 200 OK
Server: nginx/1.10.3 (Ubuntu)
Date: Thu, 01 Oct 2026 10:41:43 GMT
Content-Type: text/html
Content-Length: 483
Last-Modified: Tue, 29 Sep 2026 21:02:07 GMT
Connection: keep-alive
ETag: "6abc274f-1e3"
Cache-Control: no-store
Accept-Ranges: bytes

<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="shortcut icon" href="/icons/favicon-red.ico"/>
  <link rel="apple-touch-icon" href="/icons/icon-red.png"/>
  <title>http.badssl.com</title>
  <link rel="stylesheet" href="/style.css">
  <style>body { background: red; }</style>
</head>
<body>
<div id="content">
  <h1 style="font-size: 8vw;">
    http.badssl.com
  </h1>
</div>

</body>
</html>
root@alpha:~# curl -i -H "Host: http.badssl.com" http://outbound.k36.com
HTTP/1.1 200 OK
Server: nginx/1.10.3 (Ubuntu)
Date: Thu, 01 Oct 2026 10:43:46 GMT
Content-Type: text/html
Content-Length: 483
Last-Modified: Tue, 29 Sep 2026 21:02:07 GMT
Connection: keep-alive
ETag: "6abc274f-1e3"
Cache-Control: no-store
Accept-Ranges: bytes

<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="shortcut icon" href="/icons/favicon-red.ico"/>
  <link rel="apple-touch-icon" href="/icons/icon-red.png"/>
  <title>http.badssl.com</title>
  <link rel="stylesheet" href="/style.css">
  <style>body { background: red; }</style>
</head>
<body>
<div id="content">
  <h1 style="font-size: 8vw;">
    http.badssl.com
  </h1>
</div>

</body>
</html>
```

20.

Script untuk beberapa command soal yang bisa dilakukan pada alpha setelah melakukan restart node

[soal20.sh](#)

```
#!/bin/bash
```

```
echo "== Soal 2: internet via IP =="
```

```
ping -c 2 8.8.8.8
```

```
echo "== DNS: prab dan tedd =="
```

```
for s in 192.229.1.2 192.229.1.3; do
```

```
    echo "-- $s"
```

```
    dig +short @$s k36.com SOA
```

```
    dig +short @$s www.k36.com
```

```
    dig +short @$s alpha.k36.com TXT
```

```
    dig +short @$s -x 192.229.1.4
```

```
    dig +short @$s outbound.k36.com
```

```
    dig +noall +answer @$s abbey.k36.com
```

```
done
```

```
echo "== Soal 13: redirect =="
```

```
for u in 192.229.4.2 penny.k36.com 192.229.2.2 abbey.k36.com www.k36.com  
static.k36.com; do
```

```
    echo -n "$u -> "
```

```
    curl -s -o /dev/null -w "%{http_code} %{redirect_url}\n" http://$u
```

```
done
```

```
echo "== Soal 9 dan 10: backend lewat hostname =="
```

```
curl -s -o /dev/null -w "vault arsip: %{http_code}\n"
```

```
http://vault.k36.com/arsip/
```

```
curl -s -o /dev/null -w "core profil: %{http_code}\n" http://core.k36.com/profil
```

```
echo "== Soal 11: distribusi trafik =="
```

```
for i in 1 2 3 4 5 6; do curl -s http://www.k36.com/; echo; done
```

```
echo "== Soal 12: admin =="
```

```
curl -s -o /dev/null -w "tanpa kredensial: %{http_code}\n"
```

```
http://www.k36.com/admin/
```

```
curl -s -o /dev/null -w "dengan kredensial: %{http_code}\n" -u  
'prabs:pakar_pinter_jadi_gob***' http://www.k36.com/admin/
```

```
echo "== Soal 15 =="
```

```
curl -s http://www.k36.com/eternal/; echo
```

```
curl -s http://static.k36.com/orion/
```

```
root@alpha:~# ./soal120.sh  
== Soal 2: internet via IP ==  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=110 time=20.3 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=110 time=19.5 ms  
  
--- 8.8.8.8 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1110ms  
rtt min/avg/max/mdev = 19.450/19.889/20.329/0.439 ms  
== DNS: prab dan tedd ==  
-- 192.229.1.2  
prab.k36.com. root.k36.com. 2026100310 604800 86400 2419200 604800  
penny.k36.com.  
192.229.4.2  
"alpha"  
vault.k36.com.  
http.badssl.com.  
104.154.89.105  
abbey.k36.com. 604800 IN A 192.229.2.2  
-- 192.229.1.3  
prab.k36.com. root.k36.com. 2026100310 604800 86400 2419200 604800  
penny.k36.com.  
192.229.4.2  
"alpha"  
vault.k36.com.  
http.badssl.com.  
104.154.89.105  
abbey.k36.com. 604800 IN A 192.229.2.2  
== Soal 13: redirect ==  
192.229.4.2 -> 301 http://www.k36.com/  
penny.k36.com -> 301 http://www.k36.com/  
192.229.2.2 -> 302 http://static.k36.com/  
abbey.k36.com -> 302 http://static.k36.com/  
www.k36.com -> 200  
static.k36.com -> 200  
== Soal 9 dan 10: backend lewat hostname ==  
vault arsip: 200  
core profil: 200  
== Soal 11: distribusi trafik ==  
desmond  
  
obladi  
  
desmond  
  
obladi  
  
desmond  
  
obladi  
  
== Soal 12: admin ==  
tanpa kredensial: 401  
dengan kredensial: 404  
== Soal 15 ==  
Eternal dari penny, PHP 8.4.26  
Orion dari abbey
```